Elbow Muscle Force Optimization: Cost Function n=2 vs n=4

Cost function: $u = \sum_i (F_i / PCSA_i)^n$

Constraints: $\sum M = 0$ (moment equilibrium about elbow), $F_i \geq 0$

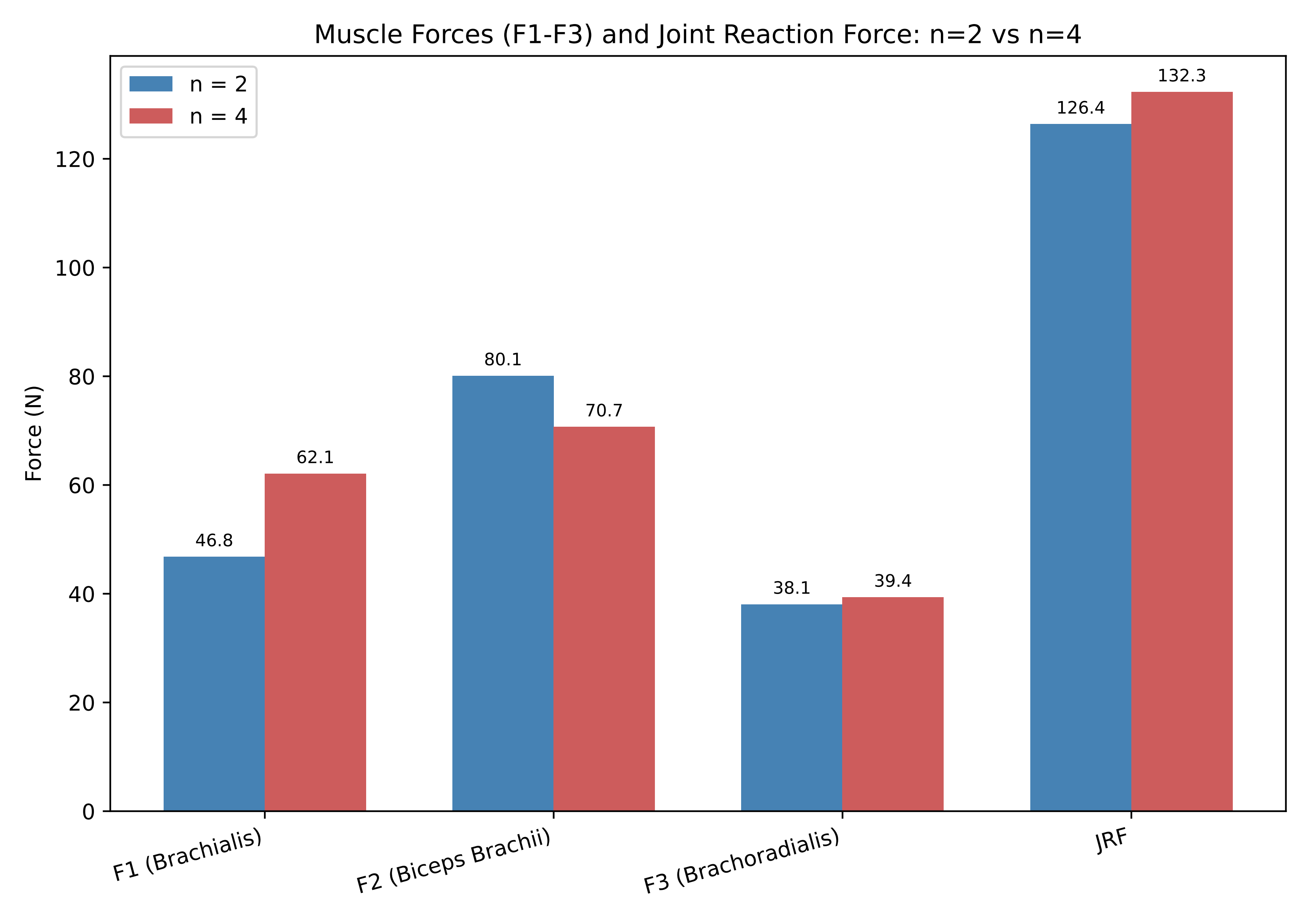
Initial conditions: $F_0 = (0, 0, 0)$ for both trials

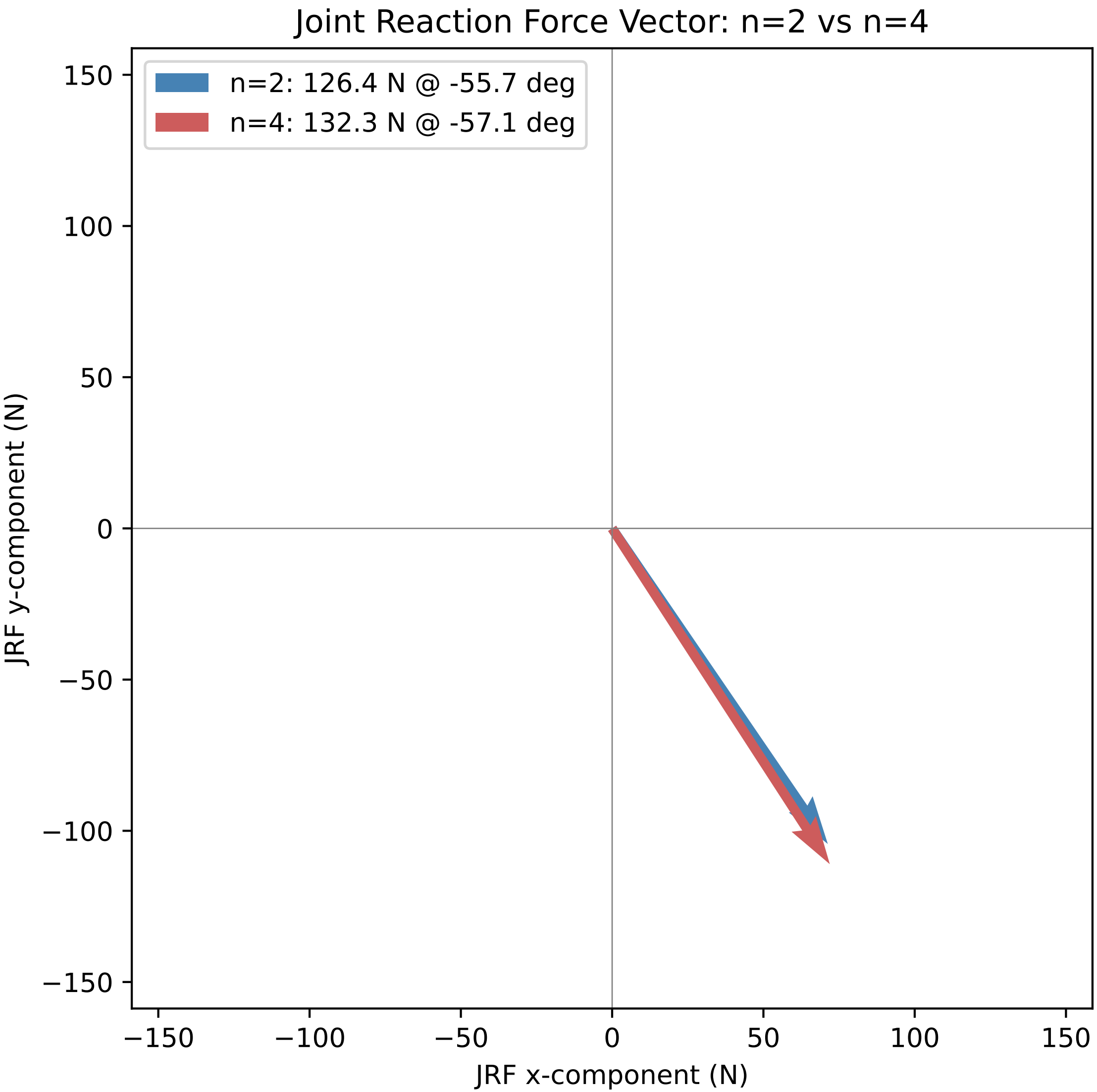
Quantity	n = 2	n = 4	Difference (n4 - n2)
F1 - Brachialis (N)	46.80	62.09	+15.29
F2 - Biceps Brachii (N)	80.09	70.69	-9.40
F3 - Brachoradialis (N)	38.06	39.35	+1.29
JRF magnitude (N)	126.39	132.31	+5.92
JRF angle (deg)	-55.67	-57.06	-1.39
Cost value, u	202.05	14647.84	n/a (not comparable across n)

Note: the cost value u is not directly comparable between n=2 and n=4 since it is raised to a different power; compare the resulting forces and JRF instead.

Interpretation

Raising the exponent n from 2 to 4 makes the cost function penalize large individual muscle stresses much more steeply. As a result, the n=4 solution redistributes load away from the muscle carrying the largest share under n=2 (Biceps Brachii, F2) and toward the muscle with the smallest moment arm (Brachialis, F1), which must work harder to help satisfy moment equilibrium. F3 (Brachoradialis) changes only slightly since it already carries a comparatively low, moderate share of the stress. Because all three muscle forces increase in aggregate contribution to the equilibrium condition, the resulting joint reaction force is slightly larger and at a slightly steeper angle under n=4 than under n=2.


Muscle Forces (F1-F3) and Joint Reaction Force: n=2 vs n=4
n = 2
n = 4
Force (N)
0
20
40
60
80
100
120
46.8
62.1
80.1
70.7
38.1
39.4
126.4
132.3
F1 (Brachialis)
F2 (Biceps Brachii)
F3 (Brachoradialis)
JRF


Joint Reaction Force Vector: n=2 vs n=4
n=2: 126.4 N @ -55.7 deg
n=4: 132.3 N @ -57.1 deg
150
100
50
0
−50
−100
−150
JRF y-component (N)
−150
−100
−50
0
50
100
150
JRF x-component (N)